

P4B Solar Mini-Grid: Socioeconomic Impact Analysis

CS 6150 – Computing for Good
Fall 2024

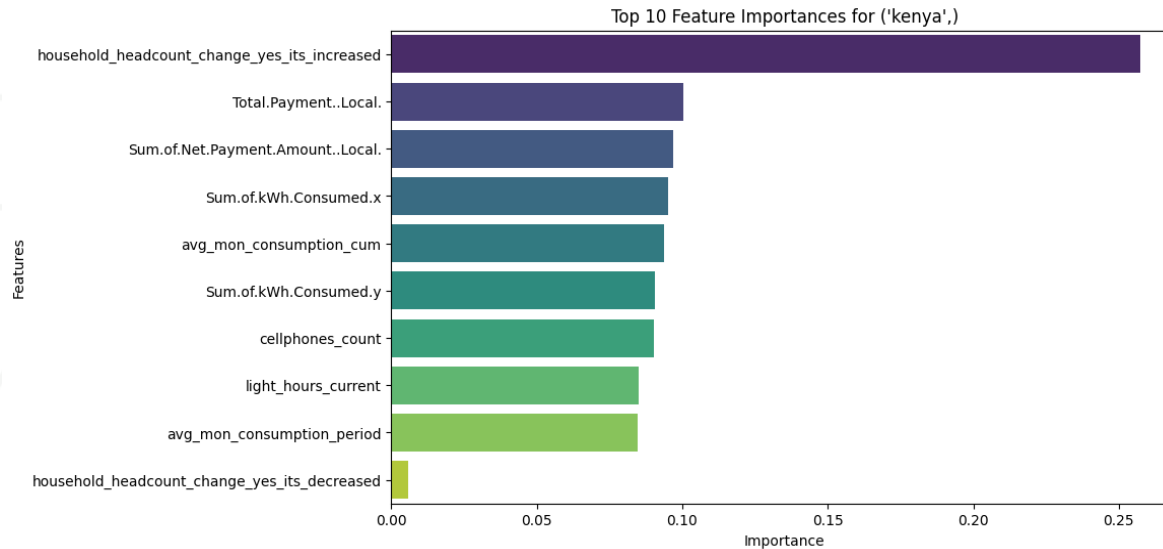
Feature Selection with RFE (Recursive Feature Elimination)

- Key selected features

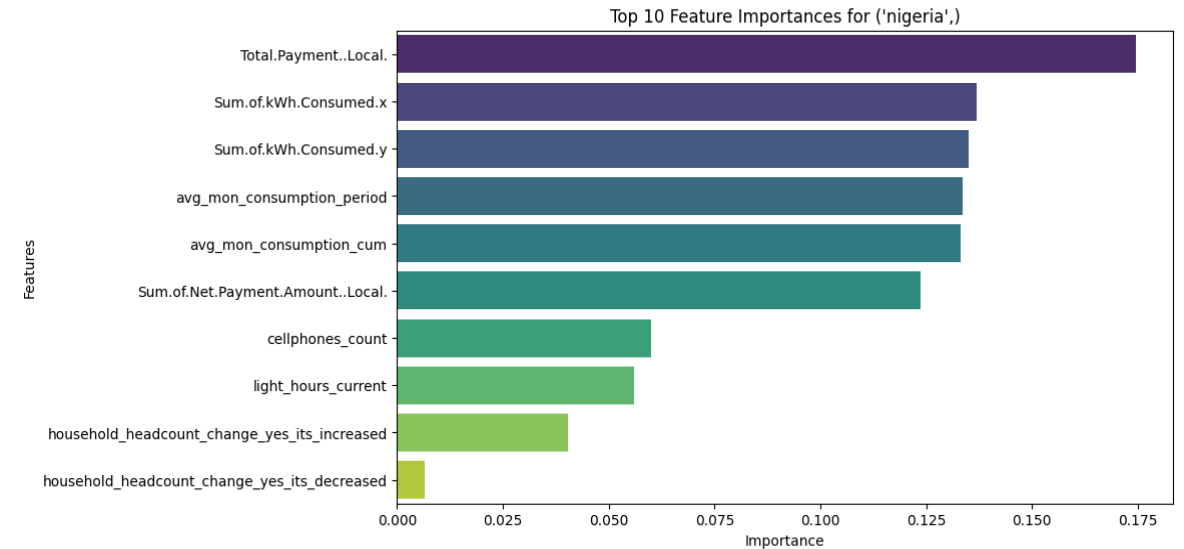
Categorical Data	Numerical Data
Household headcount change	Cellphones count
	Light hours current
	Sum.of.kwh.Consumed.x
	Total.Payment..Local
	Avg mon consumption period
	Sum.of.kwh.Consumed.y
	Sum.ofNet.Payment.Amount..Local
	Avg mom consumption cum

Random Forest Model Result for Each Country

- Kenya



- Nigeria



Model Analysis

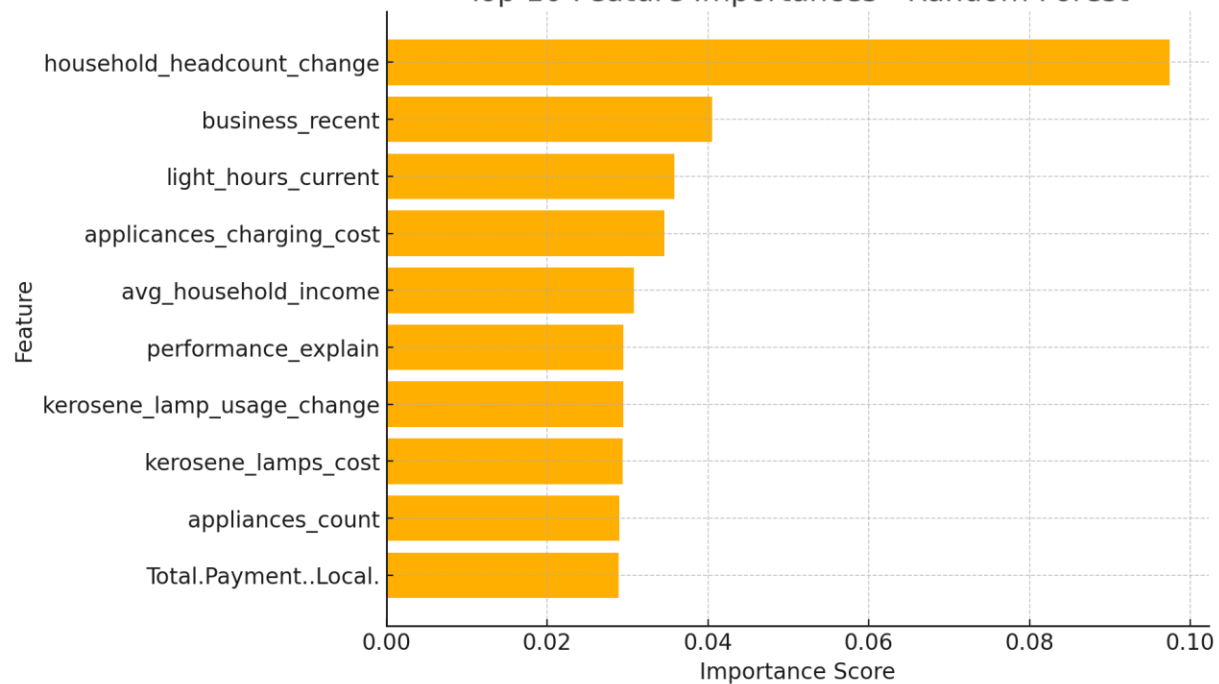
	Random Forest	Gradient Boosting	SVM	Logistic Regression (Ridge)	Logistic Regression (Lasso)
0	0.94	0.94	0.94	0.94	0.94
1	0.60	0.64	0.60	0.66	0.66
Recall	0.891	0.891	0.896	0.900	0.900
Precision	0.882	0.885	0.888	0.895	0.895
accuracy	0.89	0.89	0.90	0.90	0.90
macro avg	0.77	0.79	0.77	0.80	0.80
weighted avg	0.88	0.89	0.89	0.90	0.90

We tried 5 different models to help us to filter key driven factors to impact the dependent variable:

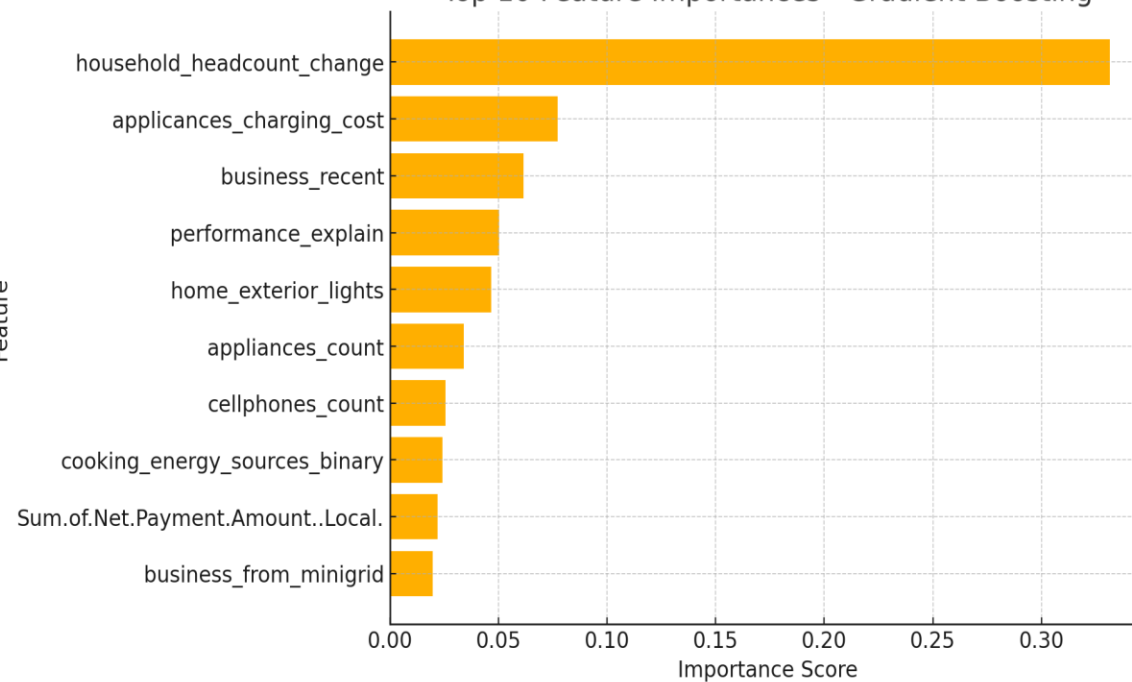
“children_school_performance_changing”

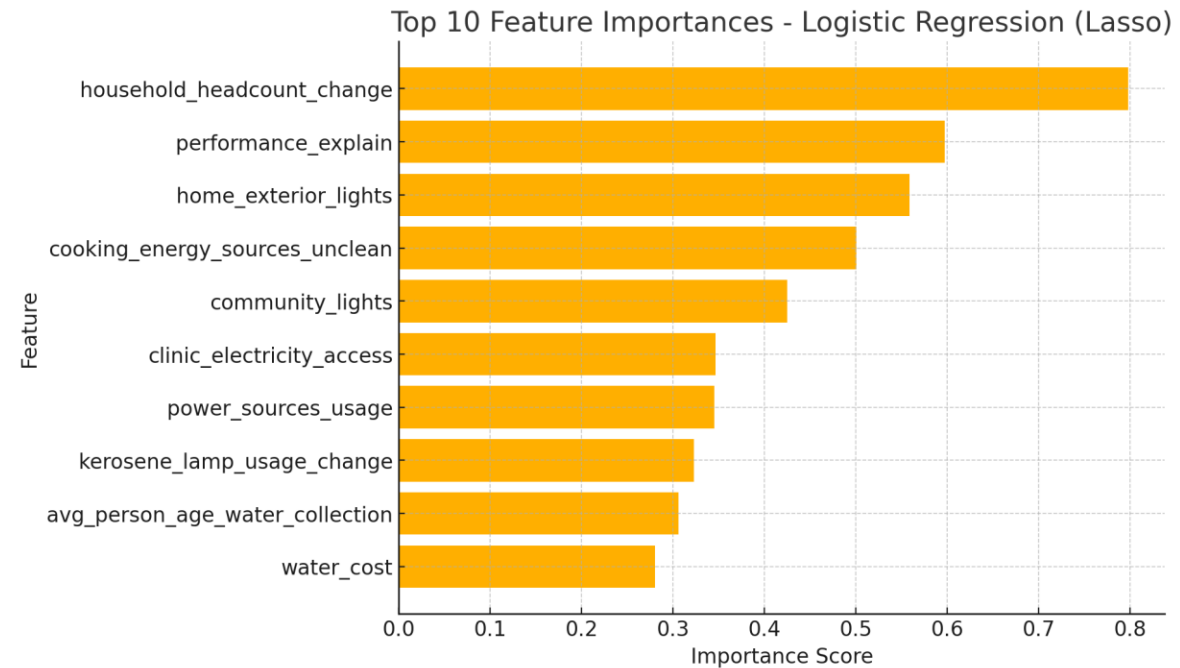
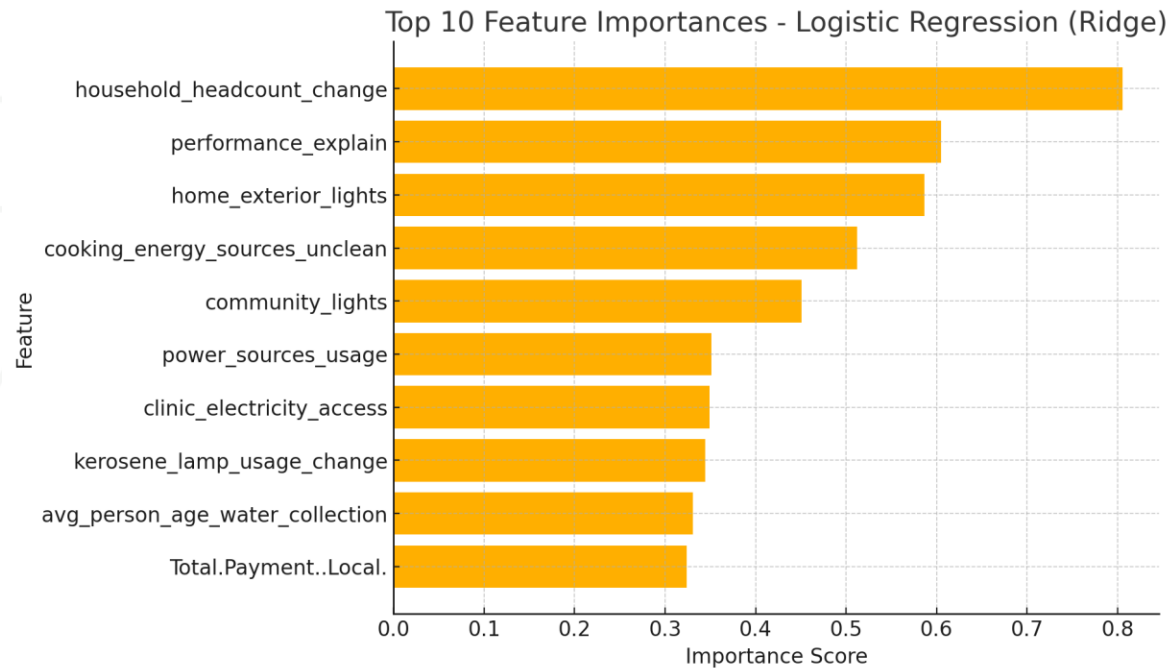
All the five models provide us with similar predicting power and performance, logistic regression is better than others.

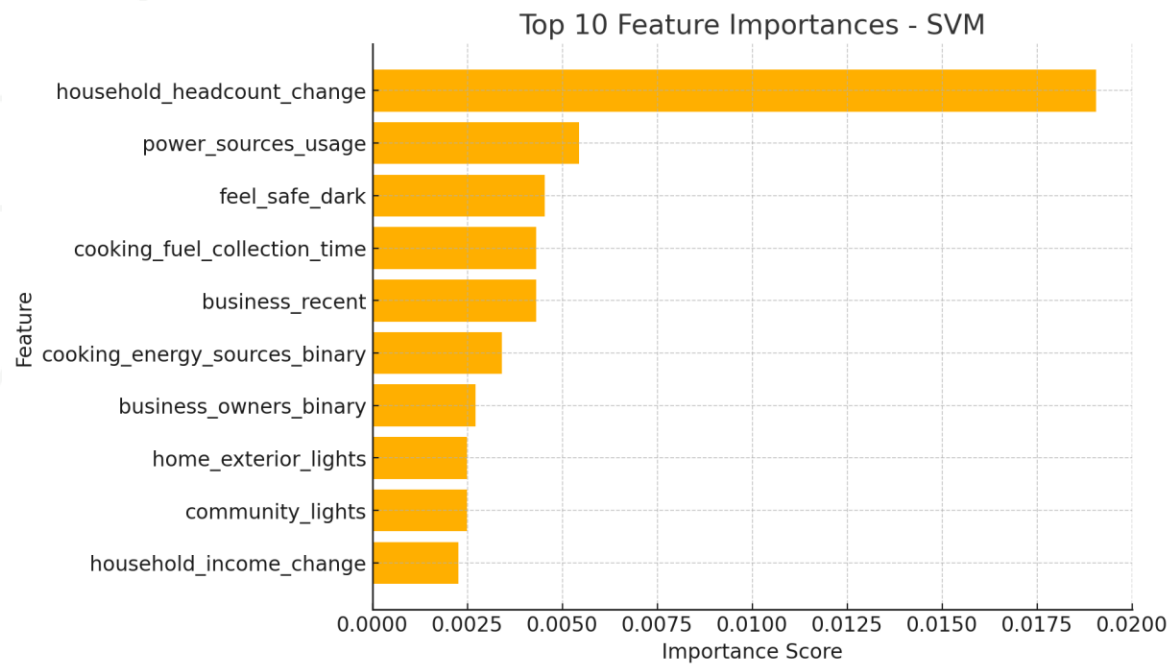
Top 10 Feature Importances - Random Forest



Top 10 Feature Importances - Gradient Boosting







Result: After reviewing the results from all the models, we decide to move forward with the following variable and test the significance level:

- Total Energy Payment (Local)
- Total Energy Consumption (kWh)
- Appliance Count
- cooking_energy_source_unclean
- home_exterior_lights
- business_recent
- household_headcount_change
- performance_explain

Statistic Result

	Test	Statistic test Value	p_value
Total_Energy_Payment_Local	Independent T-Test	5.145308	3.30E-07
Total_Energy_Consumption_kWh	Independent T-Test	0.24851	0.803835
Appliance_Count	Independent T-Test	-2.27005	0.0237414
home_exterior_lights	Chi-Square Test	119.7297	7.25E-28
business_recent	Chi-Square Test	299.9403	3.39E-67
cooking_energy_source_unclean	Chi-Square Test	107.0496	4.34E-25
community_lights	Chi-Square Test	104.3536	1.69E-24
performance_explain	Mann-Whitney UTest	158221.5	2.15E-34
household_headcount_change	Mann-Whitney UTest	327024	4.16E-76

- **Home Exterior Lights**

- **Chi-square statistic:** 119.73
- **p-value:** 0.000 (significant)
- **Interpretation:** The result indicates an association between having home exterior lights and changes in children's school performance. Households with exterior lighting may offer additional stability or security, contributing to performance changes.

- **Business Recent**

- **Chi-square statistic:** 299.94
- **p-value:** 0.000 (significant)
- **Interpretation:** This result suggests an association between recent business activity and children's performance changes. Households with recent business activity may reflect economic shifts impacting family resources and, consequently, children's school performance.

- **Cooking Energy Source (Unclean)**

- **Chi-square statistic:** 107.04
- **p-value:** 0.000 (significant)
- **Interpretation:** This result suggests an association between Cooking Energy Source and children's performance changes.

- **Community Lights**

- **Chi-square statistic:** 104.35
- **p-value:** 0.000 (significant)
- **Interpretation:** This result suggests an association between Community Lights and children's performance changes.

- **Total Energy Payment (Local)**

- **t-statistic:** 5.15
- **p-value:** 0.000 (significant)
- **Interpretation:** The significant p-value and positive t-statistic indicate that households with a change in children's school performance tend to have higher energy payments on average compared to those without a change.

- **Total Energy Consumption (kWh)**

- **t-statistic:** 0.25
- **p-value:** 0.804 (not significant)
- **Interpretation:** There is no significant difference in total energy consumption between households with and without a change in school performance. This suggests that energy consumption does not have a meaningful impact on school performance change in this sample.

- **Appliance Count**

- **t-statistic:** -2.27
- **p-value:** 0.024 (significant)
- **Interpretation:** The significant p-value and negative t-statistic indicate that households without a change in school performance tend to have a higher appliance count on average. This suggests that households with more appliances may provide a more stable environment for children.

- **Performance Explain**
 - **U-statistic:** 158221.5
 - **p-value:** 0.000 (significant)
 - **Interpretation:** This result suggests an association between Performance Explain and children's performance changes.
- **Household Headcount Change**
 - **U-statistic:** 327024
 - **p-value:** 0.000 (significant)
 - **Interpretation:** This result suggests an association between Household Headcount Change and children's performance changes.

Power and direction

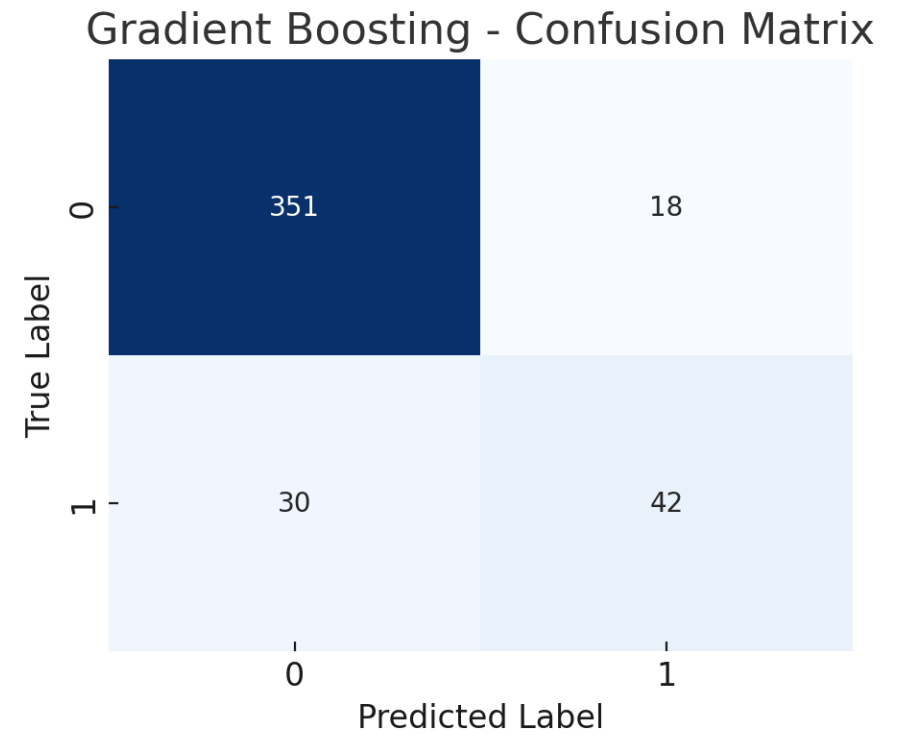
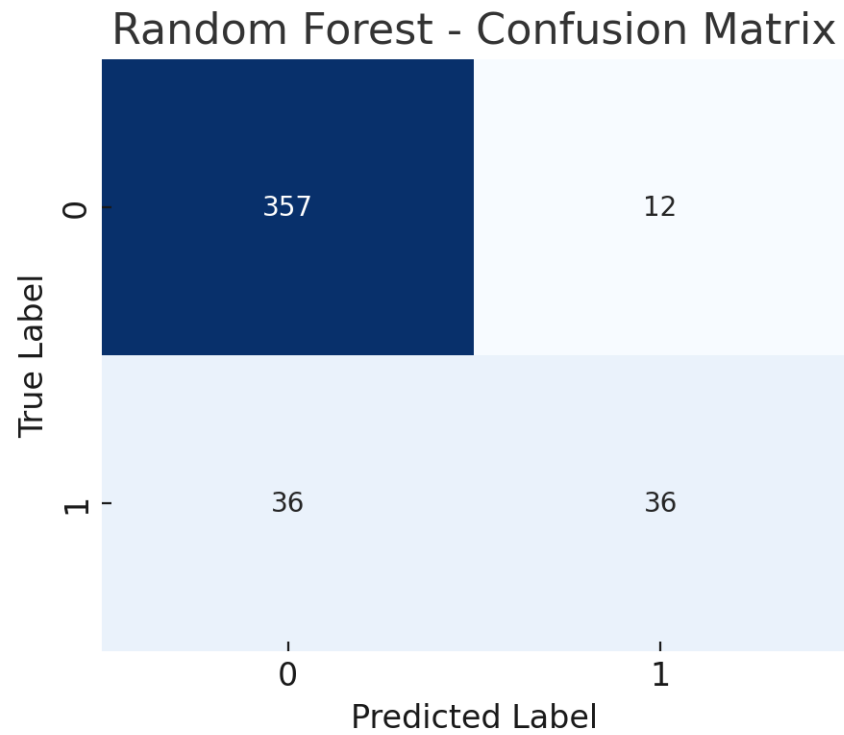
	Coef.	Std.Err.	Z-value	P> Z	[0.025	0.975]
Intercept	-4.0512	0.2340	-17.3151	0.0000	-4.5098	-3.5926
Total_Energy_Payment_Local	0.0000	0.0000	-3.1161	0.0018	0.0000	0.0000
Total_Energy_Consumption_kWh	0.0009	0.0007	1.3476	0.1778	-0.0004	0.0023
Appliance_Count	0.0340	0.0168	2.0216	0.0432	0.0010	0.0670
home_exterior_lights	1.3734	0.1900	7.2303	0.0000	1.0011	1.7457
business_recent	1.0648	0.1781	5.9783	0.0000	0.7157	1.4138
cooking_energy_source_unclean	1.3679	0.2242	6.1006	0.0000	0.9284	1.8074
community_lights	0.5356	0.1747	3.0655	0.0022	0.1932	0.8781
performance_explain	1.3495	0.1645	8.2059	0.0000	1.0272	1.6718
household_headcount_change	-2.1347	0.2005	-10.6448	0.0000	-2.5277	-1.7416

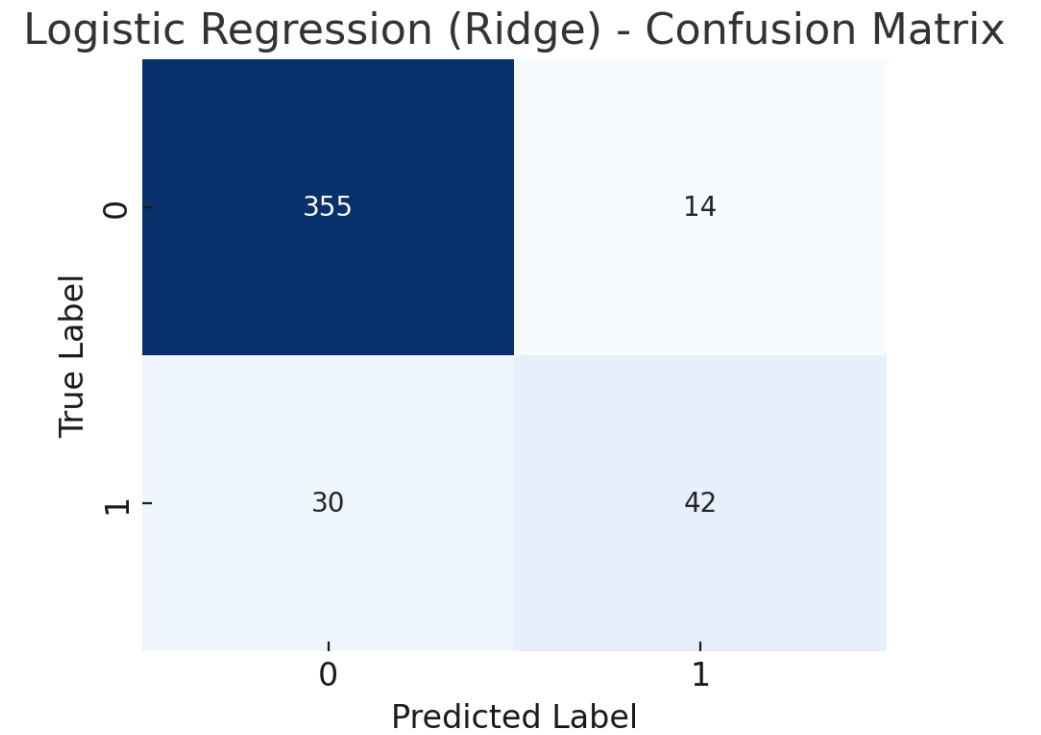
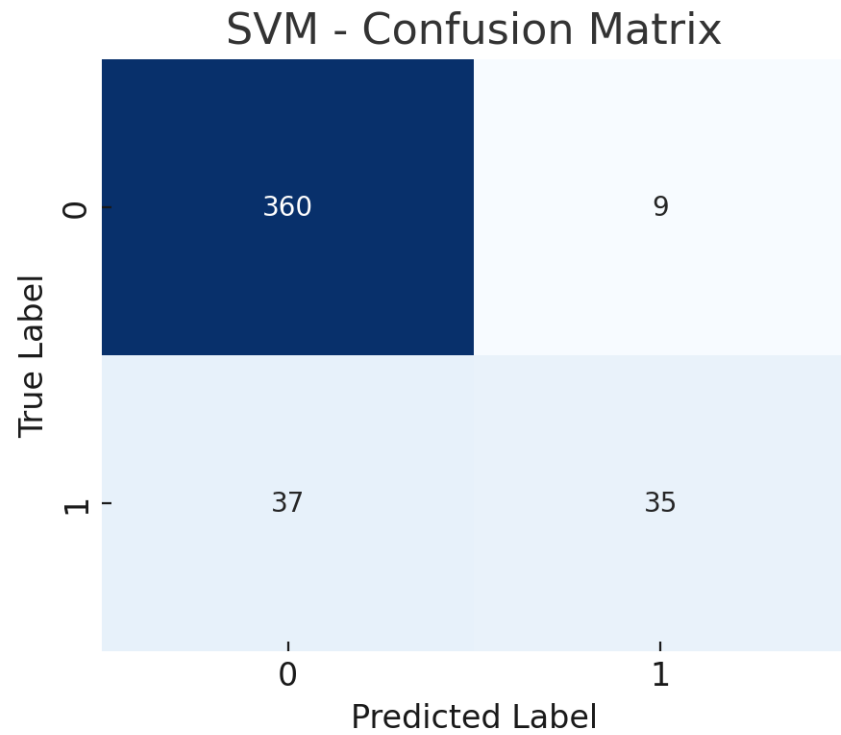
Overall Interpretation

Significant predictors in the model, such as **Home Exterior Lights**, **Business Recent**, **Cooking Energy Source (Unclean)**, and **Household Headcount Change**, highlight factors in the household environment that influence school performance. Economic stability (business activity), home security (exterior lights), and household stability (headcount) all play substantial roles in supporting children's education outcomes

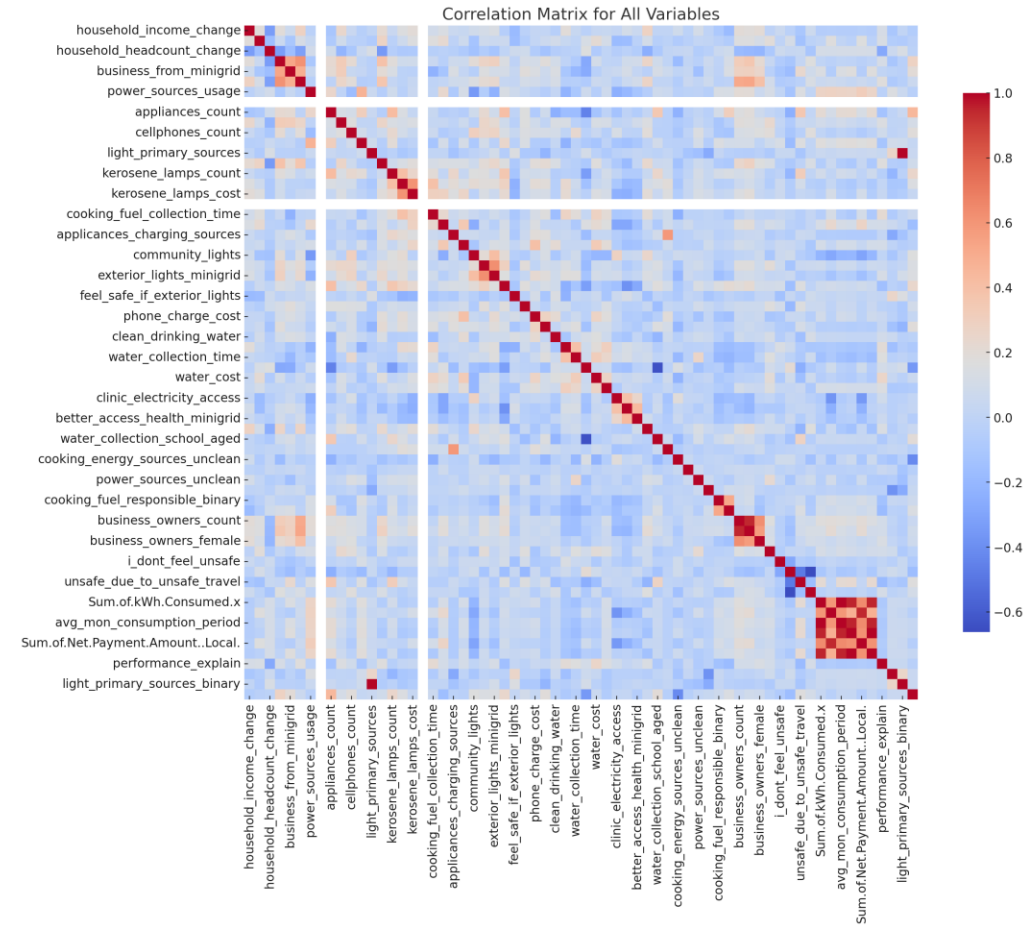
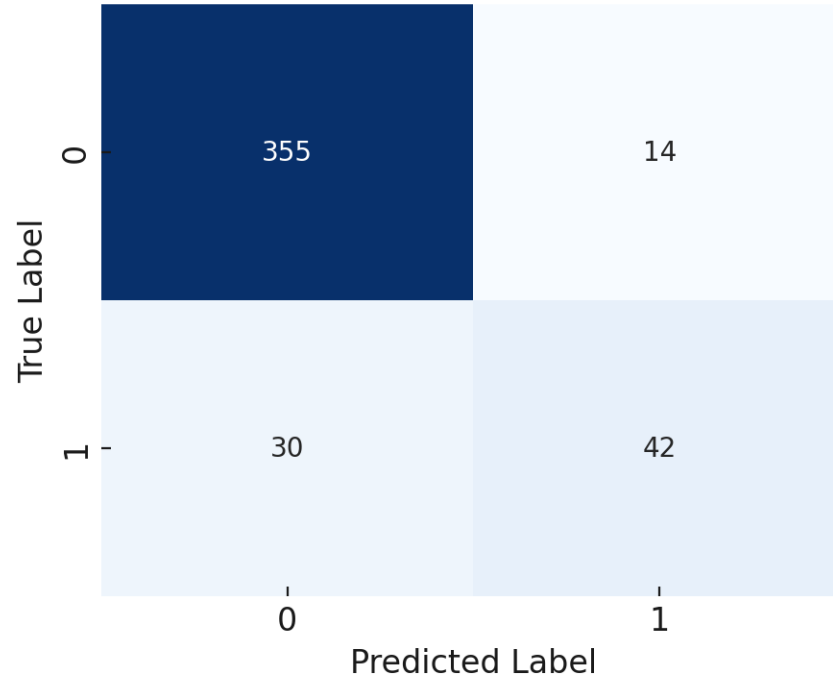
- **Intercept**
 - **Coefficient:** -4.0512
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** The negative intercept suggests a low baseline likelihood of a change in children's school performance when all predictors are at zero.
- **Total Energy Payment (Local)**
 - **Coefficient:** -0.0000
 - **p-value:** 0.0018 (significant)
 - **Interpretation:** There is a small but significant inverse relationship, suggesting that higher energy payments are slightly associated with reduced odds of a school performance change.
- **Total Energy Consumption (kWh)**
 - **Coefficient:** 0.0009
 - **p-value:** 0.1778 (not significant)
 - **Interpretation:** This positive coefficient is not statistically significant, indicating no meaningful association between energy consumption and changes in school performance.
- **Appliance Count**
 - **Coefficient:** 0.0340
 - **p-value:** 0.0432 (significant)
 - **Interpretation:** A positive association, suggesting that each additional appliance slightly increases the likelihood of a performance change. More appliances might reflect better home conditions, which could support children's education.
- **Home Exterior Lights**
 - **Coefficient:** 1.3734
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** Strongly associated with increased odds of school performance change, possibly indicating that households with exterior lights (a proxy for security or stability) support better educational outcomes.
- **Business Recent**
 - **Coefficient:** 1.0648
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** Recent business activity in the household is associated with a higher likelihood of performance change, suggesting economic activity may positively impact the household environment for education.
- **Cooking Energy Source (Unclean)**
 - **Coefficient:** 1.3679
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** A significant positive relationship, suggesting that unclean cooking energy sources are associated with a higher likelihood of school performance change, possibly reflecting households under economic or health strain.
- **Community Lights**
 - **Coefficient:** 0.5356
 - **p-value:** 0.0022 (significant)
 - **Interpretation:** This positive relationship suggests that community lighting is associated with improved odds of school performance change, likely due to better study environments.
- **Performance Explain**
 - **Coefficient:** 1.3495
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** A strong positive association indicates that households with higher performance-explain scores (satisfaction or expectations) are more likely to experience school performance changes.
- **Household Headcount Change**
 - **Coefficient:** -2.1347
 - **p-value:** 0.0000 (highly significant)
 - **Interpretation:** A negative association suggests that households with headcount changes (possibly indicating instability) are less likely to see positive school performance changes.

Appendix





Logistic Regression (Lasso) - Confusion Matrix



Distribution of Key Variables by Children's School Performance Change

